

Practical Mini Project

Applying an improving the efficiency of a school management system

1. The School Management System (SMS) serves as a unified platform that integrates administrative, academic, and financial operations within an educational institution. It connects internal users such as students, teachers, and administrators with external parties like parents and payment platforms. The system manages key functions including enrollment, scheduling, grading, fee processing, and communication. It processes inputs such as student records, course details, and payment data through validation, allocation, and evaluation workflows to produce transcripts, timetables, financial reports, and performance analytics.

2. System Mapping (Causal Loop Diagram Description)

Using a Causal Loop Diagram (CLD) approach:

Reinforcing Loop (R1: Efficiency Loop)

- Improved system automation → faster processes → better user satisfaction → increased system usage → more accurate data → improved automation

Balancing Loop (B1: Resource Constraint Loop)

- Increased enrollment → higher demand for classes → resource strain (teachers/classrooms) → scheduling adjustments → limits on enrollment

Flow Relationships

- Enrollment → Scheduling → Teaching → Grading → Reporting → Feedback to students

This mapping shows how changes in one subsystem affect the entire system, emphasizing interdependence.

3. Identify System Issues

Key Bottlenecks

- Delays in data entry affecting all downstream processes
- Overloaded scheduling system during peak enrollment
- Communication delays between departments

Feedback Loop Issues

- Poor data input → inaccurate reports → poor decisions → further inefficiencies
- High enrollment without scaling resources → reduced quality of education

4. Analyze Root Causes (5 Whys Example)

Problem: Inaccurate Student Records

1. Why? → Incorrect data entry
2. Why? → Manual input errors
3. Why? → Lack of validation tools
4. Why? → Poor system design

Improvement Area	Key Features
Automation	Automated data validation during entry; auto-generation of timetables and reports.
Workflow Redesign	Streamlined enrollment process with fewer manual steps; integration between academic and financial systems
Communication Enhancements	Real-time notifications (SMS/email/app alerts); centralized communication dashboard.
Data Integration	Unified database to eliminate duplication; API integrations for external services (e.g., payments)

6. Implementation Considerations

Challenges

- Resistance to change from staff
- Training requirements for users
- Initial cost of system deployment

Resources Needed

- Technical infrastructure (servers, software)
- Skilled IT personnel
- Training programs for stakeholders

Stakeholder Impact

- Students: Improved access to information
- Teachers: Reduced administrative workload
- Administrators: Better decision-making tools